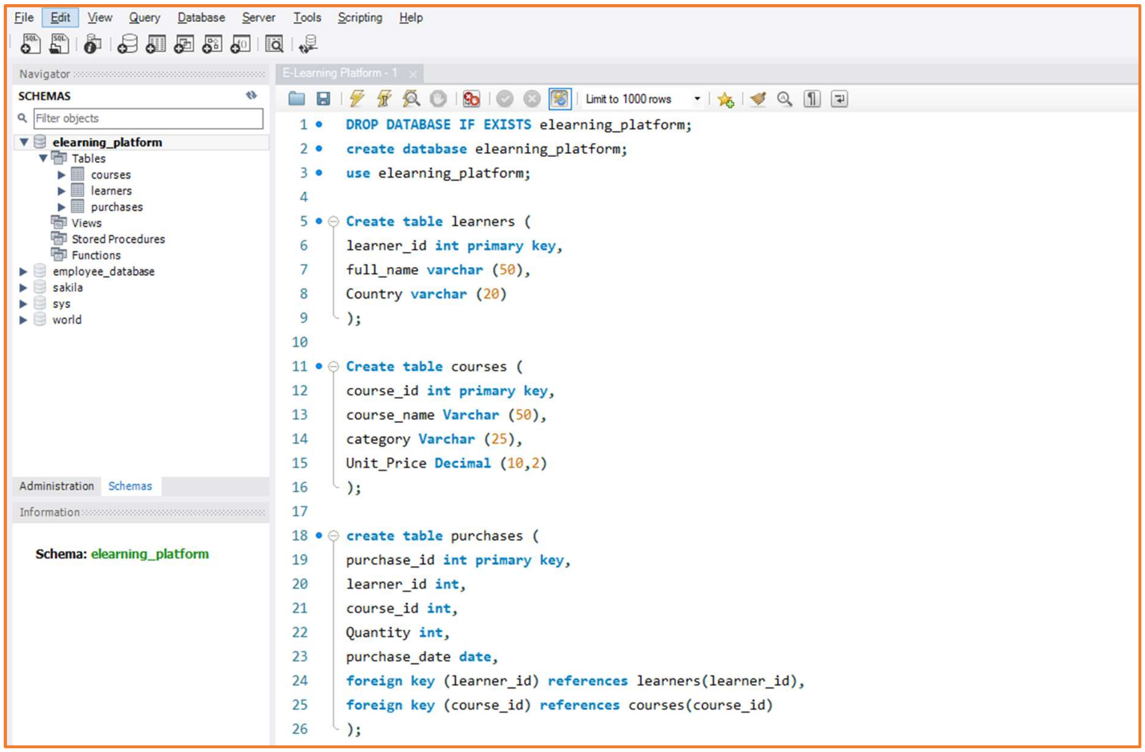
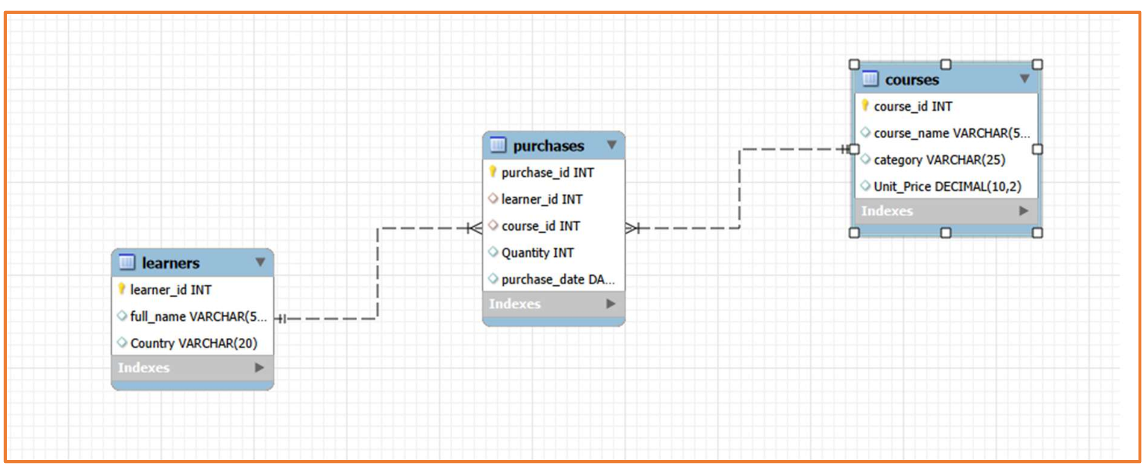
“Analysing E-Learning Platform Purchases using MySQL”

Tasks:

1. Create the database and schema. Populate the Schema:
    - Create a Database for this project and
    - Create all three tables in MySQL with appropriate data types and relationships.



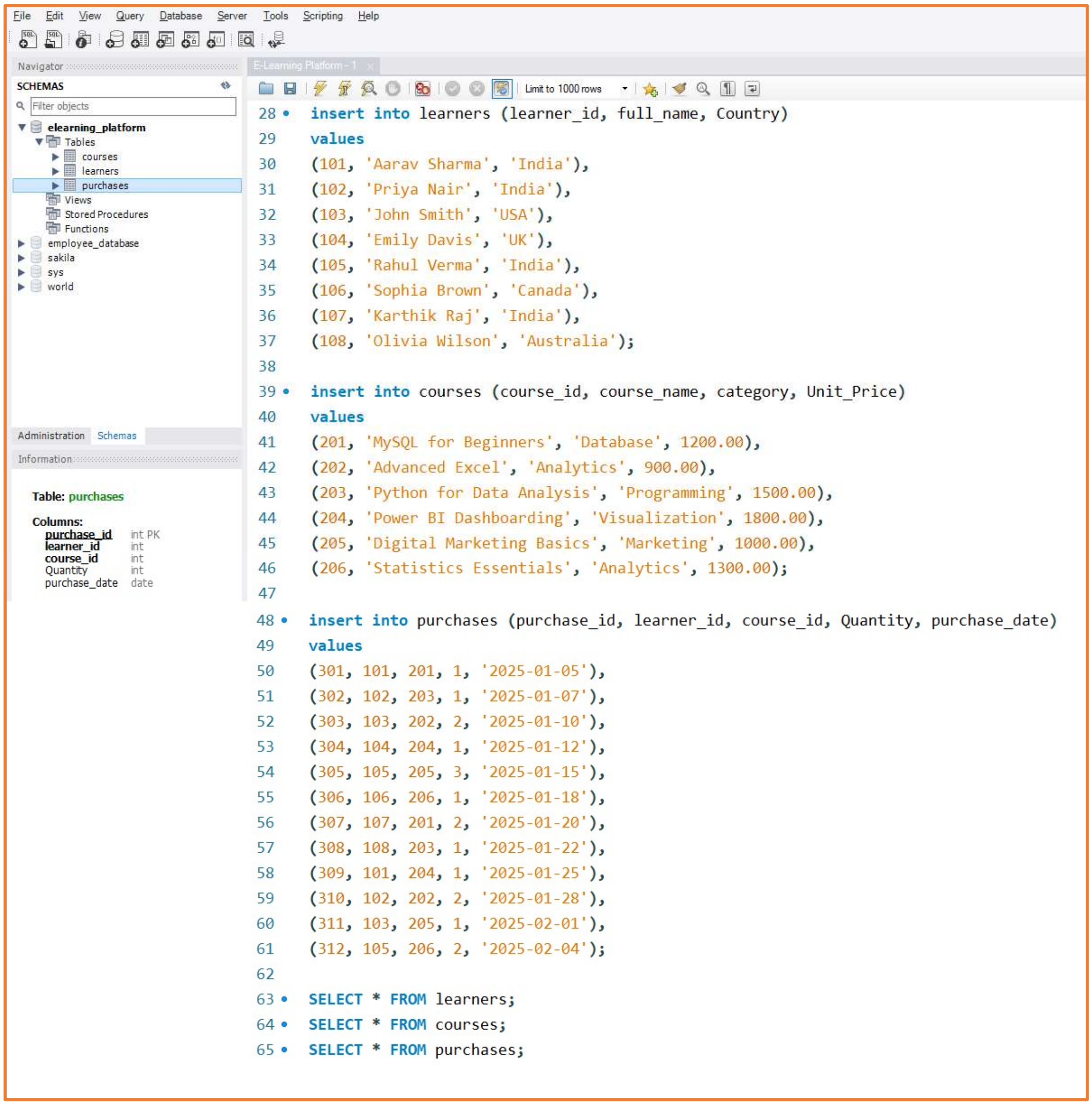


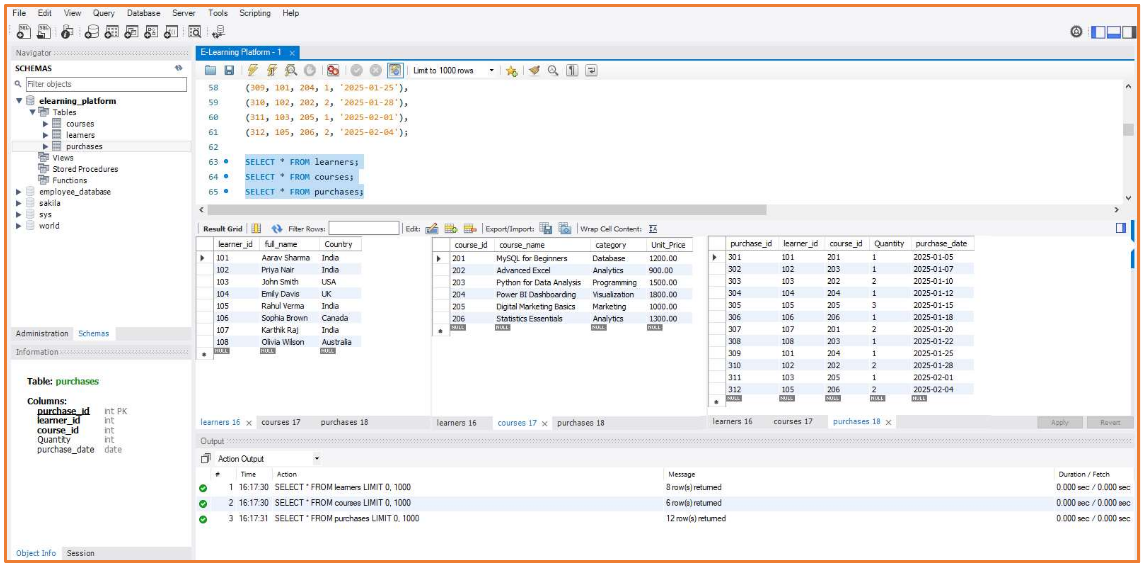
● Insert sample data covering at least:

○ 4–5 learners

○ 4–5 courses (spread across multiple categories)

○ 6–8 purchase records

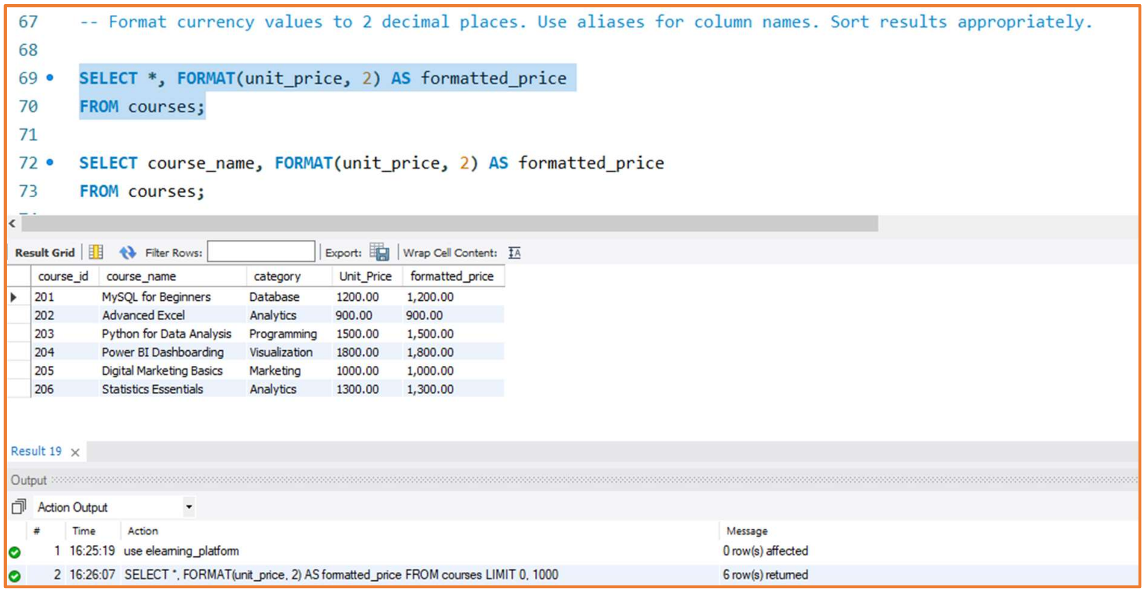


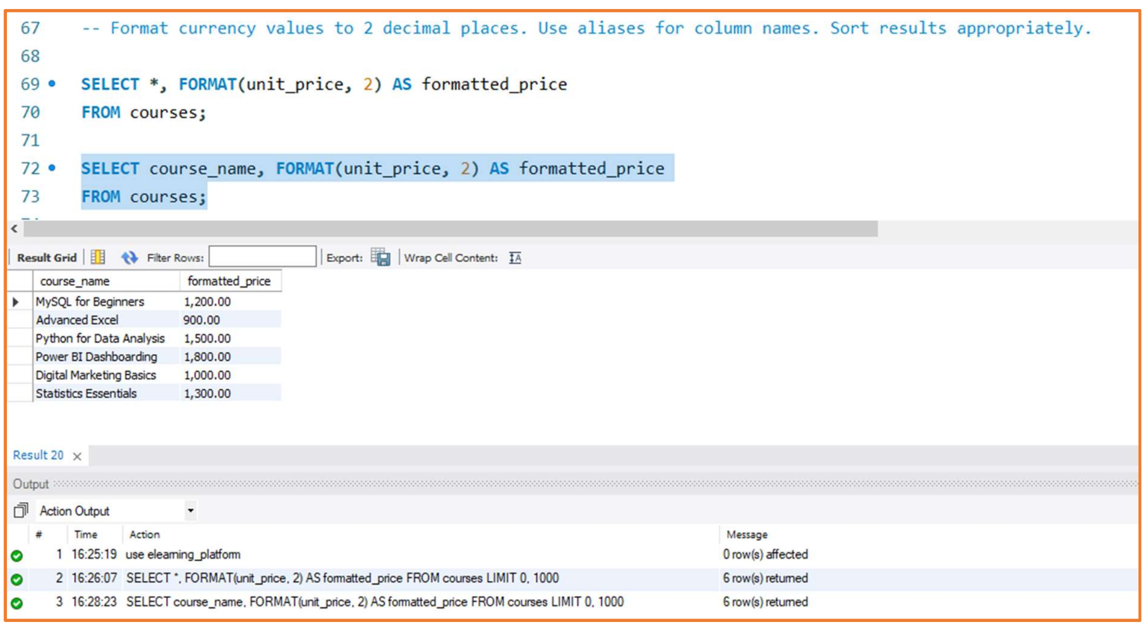

E-Learning Platform - 1
Limit to 1000 rows
(309, 101, 204, 1, '2025-01-25'),
(310, 102, 202, 2, '2025-01-28'),
(311, 103, 205, 1, '2025-02-01'),
(312, 105, 206, 2, '2025-02-04');
SELECT * FROM learners;
SELECT * FROM courses;
SELECT * FROM purchases;
Result Grid
learner_id full_name Country
101 Aarav Sharma India
102 Priya Nair India
103 John Smith USA
104 Emily Davis UK
105 Rahul Verma India
106 Sophia Brown Canada
107 Karthik Raj India
108 Olivia Wilson Australia
course_id course_name category Unit_Price
201 MySQL for Beginners Database 1200.00
202 Advanced Excel Analytics 900.00
203 Python for Data Analysis Programming 1500.00
204 Power BI Dashboarding Visualization 1800.00
205 Digital Marketing Basics Marketing 1000.00
206 Statistics Essentials Analytics 1300.00
purchase_id learner_id course_id Quantity purchase_date
301 101 201 1 2025-01-05
302 102 203 1 2025-01-07
303 103 202 2 2025-01-10
304 104 204 1 2025-01-12
305 105 205 3 2025-01-15
306 106 206 1 2025-01-18
307 107 201 2 2025-01-20
308 108 203 1 2025-01-22
309 101 204 1 2025-01-25
310 102 202 2 2025-01-28
311 103 205 1 2025-02-01
312 105 206 2 2025-02-04
learners 16
courses 17
purchases 18
Output
Action Output
1 16:17:30 SELECT * FROM learners LIMIT 0, 1000 8 row(s) returned 0.000 sec / 0.000 sec
2 16:17:30 SELECT * FROM courses LIMIT 0, 1000 6 row(s) returned 0.000 sec / 0.000 sec
3 16:17:31 SELECT * FROM purchases LIMIT 0, 1000 12 row(s) returned 0.000 sec / 0.000 sec
Table: purchases
Columns:
purchase_id int PK
learner_id int
course_id int
Quantity int
purchase_date date

2. Data Exploration Using Joins

Data Presentation Guidelines for the following query

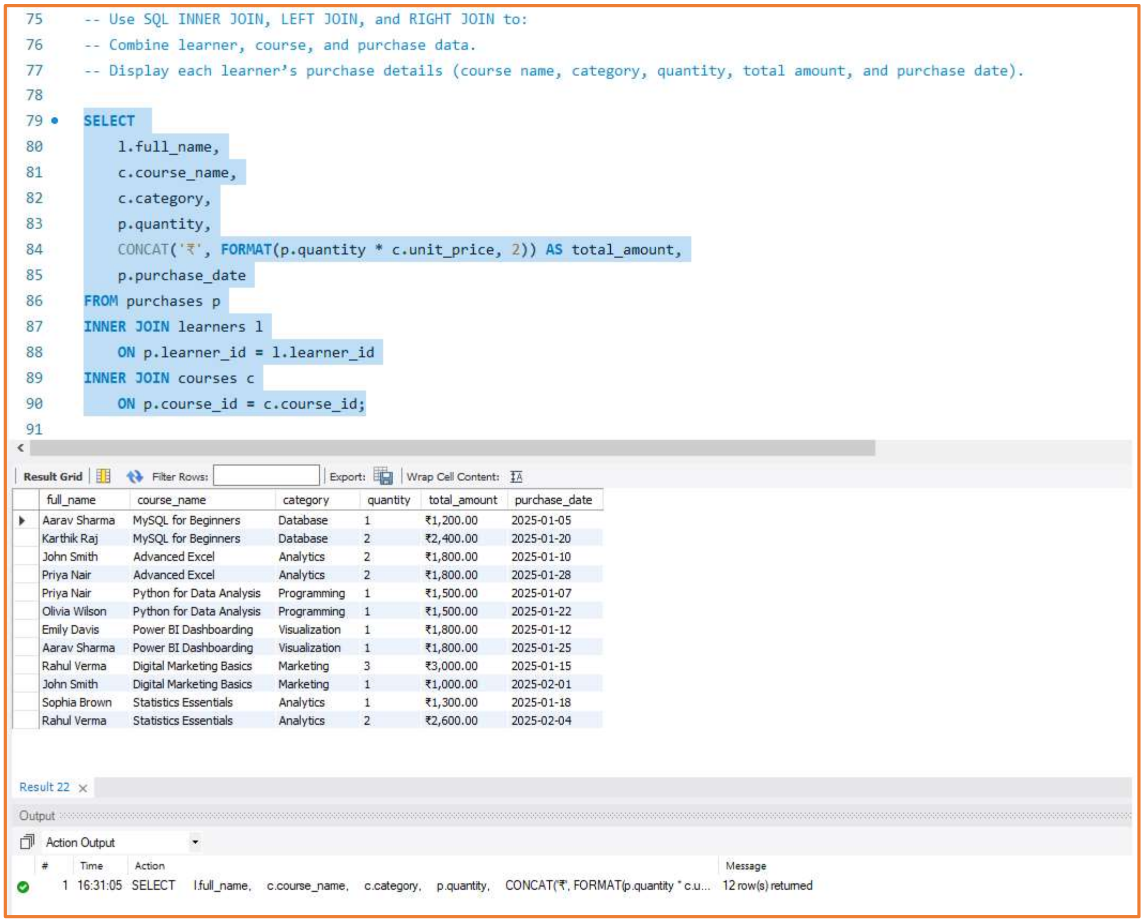
- Format currency values to 2 decimal places.
- Use aliases for column names (e.g., AS total_revenue).
- Sort results appropriately (e.g., highest total_spent first).





Use SQL INNER JOIN, LEFT JOIN, and RIGHT JOIN to:

● Combine learner, course, and purchase data.
● Display each learner’s purchase details (course name, category, quantity, total amount, and purchase date).



```sql
-- Use SQL INNER JOIN, LEFT JOIN, and RIGHT JOIN to:
-- Combine learner, course, and purchase data.
-- Display each learner’s purchase details (course name, category, quantity, total amount, and purchase date).

SELECT
    l.full_name,
    c.course_name,
    c.category,
    p.quantity,
    CONCAT('₹', FORMAT(p.quantity * c.unit_price, 2)) AS total_amount,
    p.purchase_date
FROM purchases p
INNER JOIN learners l
    ON p.learner_id = l.learner_id
INNER JOIN courses c
    ON p.course_id = c.course_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

full_name	course_name	category	quantity	total_amount	purchase_date
Aarav Sharma	MySQL for Beginners	Database	1	₹1,200.00	2025-01-05
Karthik Raj	MySQL for Beginners	Database	2	₹2,400.00	2025-01-20
John Smith	Advanced Excel	Analytics	2	₹1,800.00	2025-01-10
Priya Nair	Advanced Excel	Analytics	2	₹1,800.00	2025-01-28
Priya Nair	Python for Data Analysis	Programming	1	₹1,500.00	2025-01-07
Olivia Wilson	Python for Data Analysis	Programming	1	₹1,500.00	2025-01-22
Emily Davis	Power BI Dashboarding	Visualization	1	₹1,800.00	2025-01-12
Aarav Sharma	Power BI Dashboarding	Visualization	1	₹1,800.00	2025-01-25
Rahul Verma	Digital Marketing Basics	Marketing	3	₹3,000.00	2025-01-15
John Smith	Digital Marketing Basics	Marketing	1	₹1,000.00	2025-02-01
Sophia Brown	Statistics Essentials	Analytics	1	₹1,300.00	2025-01-18
Rahul Verma	Statistics Essentials	Analytics	2	₹2,600.00	2025-02-04

Result 22 ×

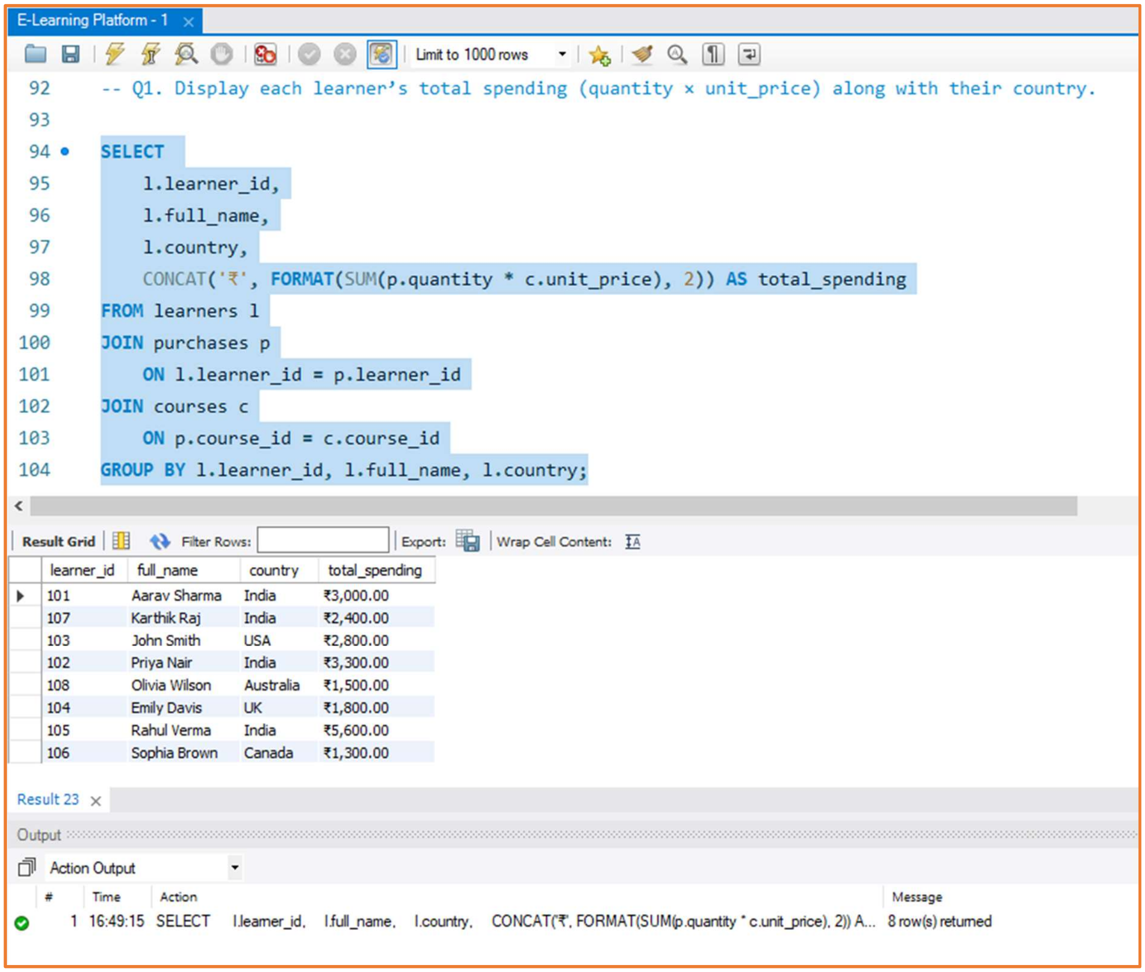
Output

Action Output

#	Time	Action	Message
1	16:31:05	SELECT l.full_name, c.course_name, c.category, p.quantity, CONCAT('₹', FORMAT(p.quantity * c.u...	12 row(s) returned

3. Analytical Queries:

Q1. Display each learner's total spending (quantity × unit_price) along with their country.



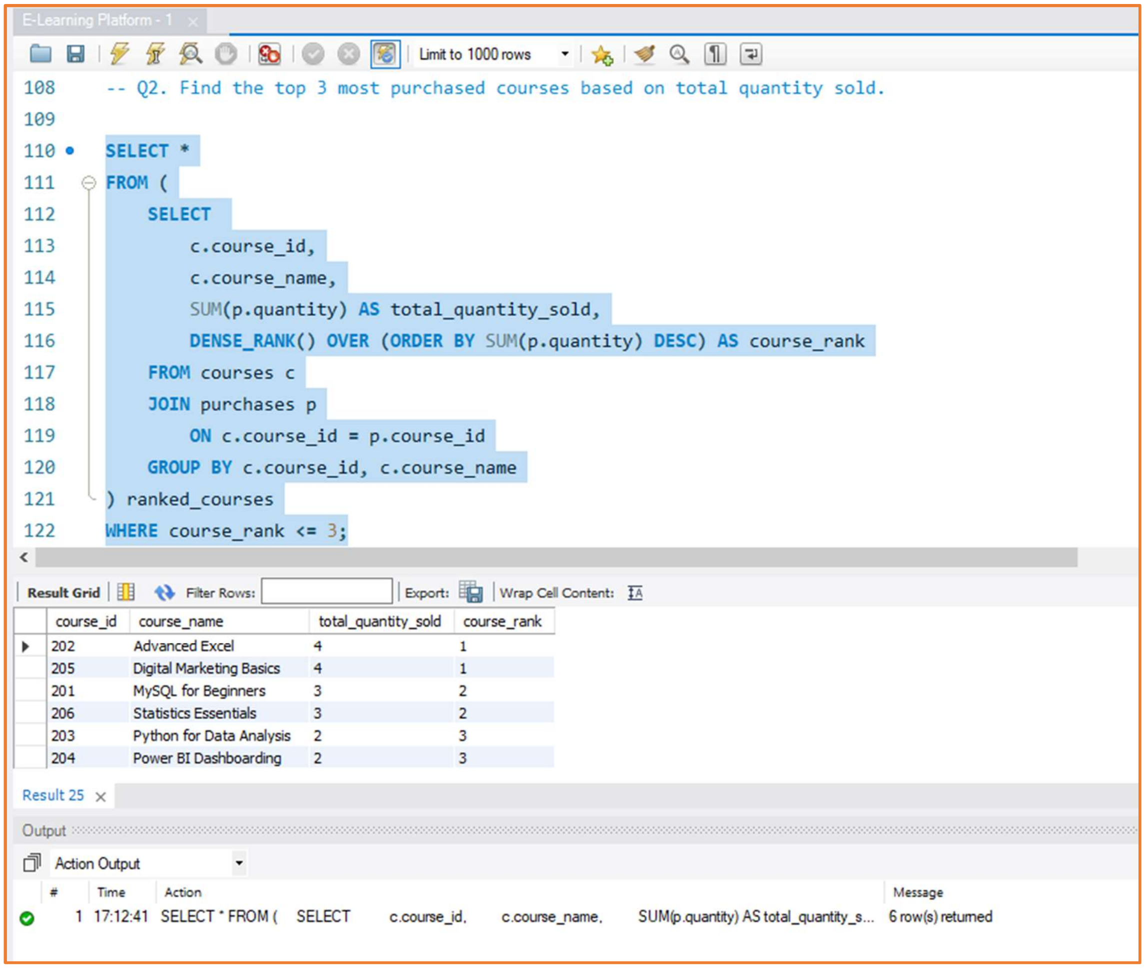
```sql
-- Q1. Display each learner's total spending (quantity × unit_price) along with their country.

SELECT
    l.learner_id,
    l.full_name,
    l.country,
    CONCAT('₹', FORMAT(SUM(p.quantity * c.unit_price), 2)) AS total_spending
FROM learners l
JOIN purchases p
    ON l.learner_id = p.learner_id
JOIN courses c
    ON p.course_id = c.course_id
GROUP BY l.learner_id, l.full_name, l.country;
```

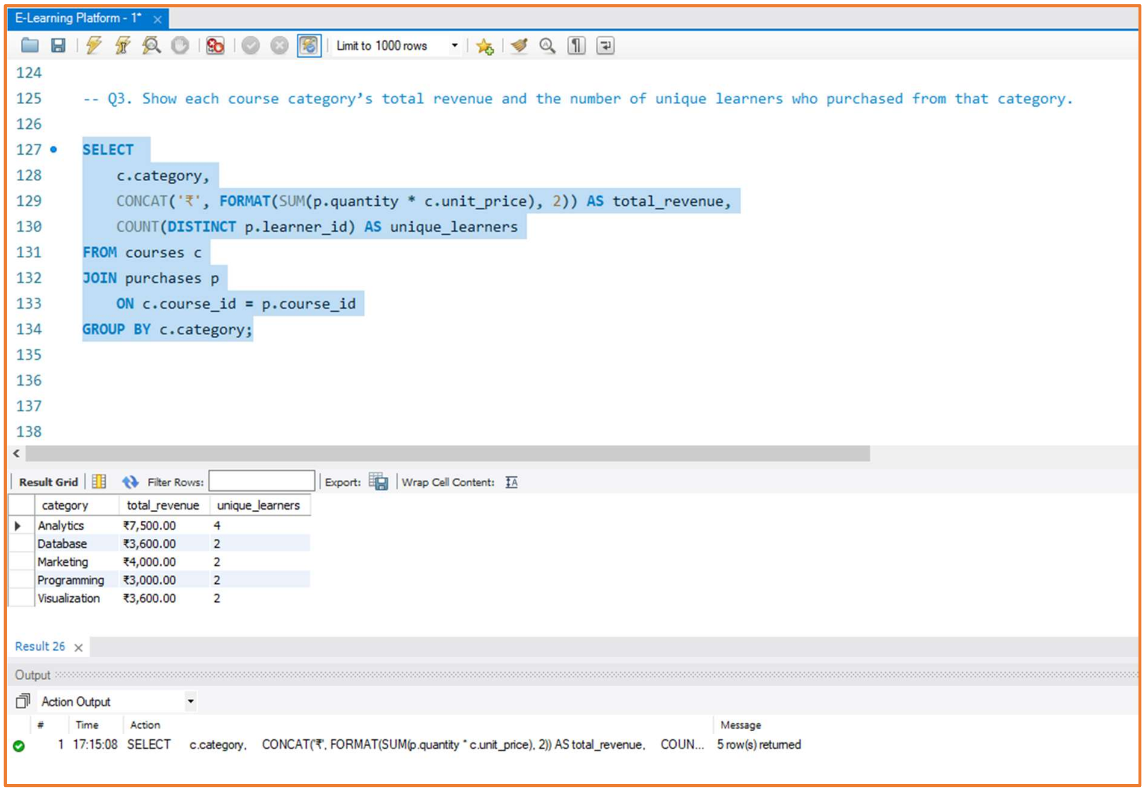
learner_id	full_name	country	total_spending
101	Aarav Sharma	India	₹3,000.00
107	Karthik Raj	India	₹2,400.00
103	John Smith	USA	₹2,800.00
102	Priya Nair	India	₹3,300.00
108	Olivia Wilson	Australia	₹1,500.00
104	Emily Davis	UK	₹1,800.00
105	Rahul Verma	India	₹5,600.00
106	Sophia Brown	Canada	₹1,300.00

#	Time	Action	Message
1	16:49:15	SELECT l.learner_id, l.full_name, l.country, CONCAT('₹', FORMAT(SUM(p.quantity * c.unit_price), 2)) A...	8 row(s) returned

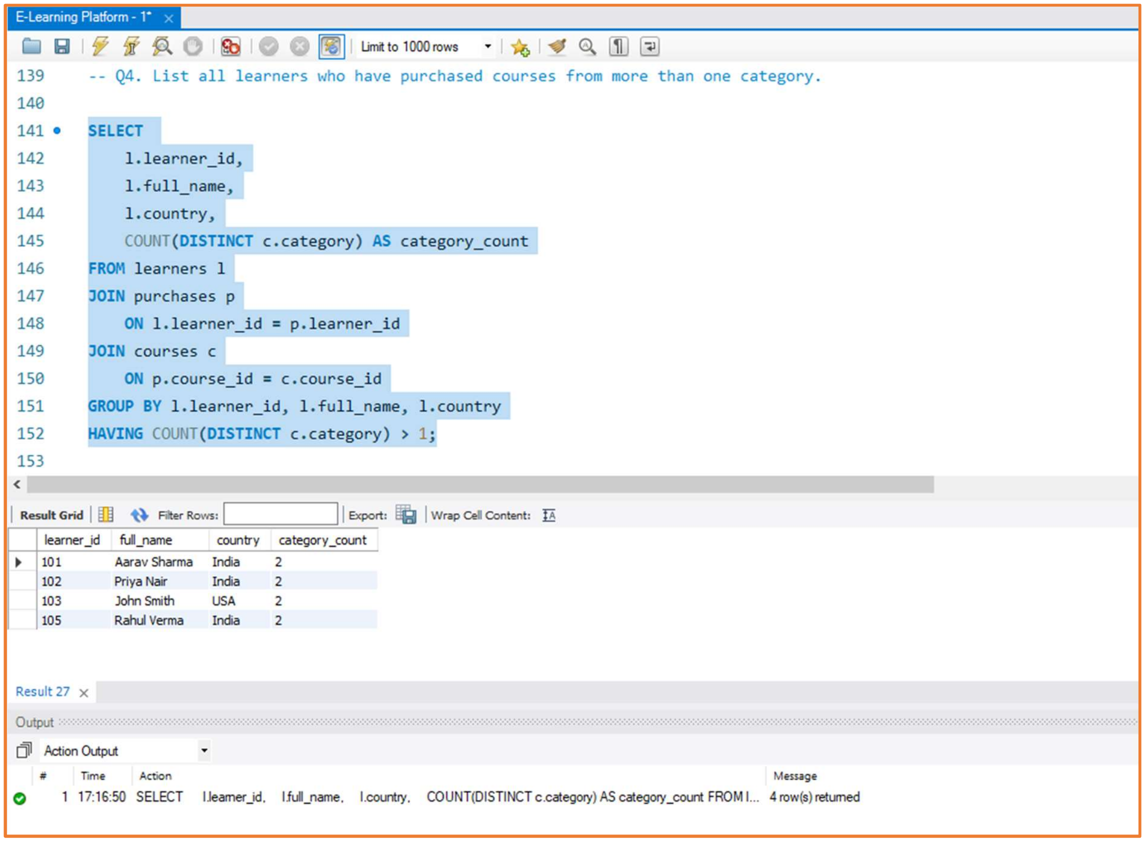
Q2. Find the top 3 most purchased courses based on total quantity sold.



Q3. Show each course category's total revenue and the number of unique learners who purchased from that category.



Q4. List all learners who have purchased courses from more than one category.



E-Learning Platform - 1*

Limit to 1000 rows

```sql
-- Q4. List all learners who have purchased courses from more than one category.

SELECT
    l.learner_id,
    l.full_name,
    l.country,
    COUNT(DISTINCT c.category) AS category_count
FROM learners l
JOIN purchases p
    ON l.learner_id = p.learner_id
JOIN courses c
    ON p.course_id = c.course_id
GROUP BY l.learner_id, l.full_name, l.country
HAVING COUNT(DISTINCT c.category) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

learner_id	full_name	country	category_count
101	Aarav Sharma	India	2
102	Priya Nair	India	2
103	John Smith	USA	2
105	Rahul Verma	India	2

Result 27

Output

Action Output

#	Time	Action	Message
1	17:16:50	SELECT l.learner_id, l.full_name, l.country, COUNT(DISTINCT c.category) AS category_count FROM l...	4 row(s) returned

Q5. Identify courses that have not been purchased at all.